

Serhiy Tolochko

Senior Backend / Platform Engineer

Python • Backend Platforms • Integrations

Background in Systems & Security Engineering

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Professional Summary

Senior Software Engineer with 15+ years of experience designing and building scalable backend platforms, automation systems, and integration-heavy business applications.

Specialized in Python backend engineering (Flask/FastAPI), API architecture, asynchronous processing, workflow orchestration, and operational automation for e-commerce and SaaS platforms.

Strong background in system-level C/C++ development, security-critical software, cryptography, and TrustZone-based environments. Proven track record of modernizing legacy systems, evolving large operational platforms, and building resilient architectures across backend, integrations, asynchronous workflows, and frontend modernization initiatives.

Core Expertise

- Backend Architecture & API Design
- Distributed Integrations & Automation
- E-commerce & Marketplace Platforms
- Workflow Automation & Async Processing
- Legacy System Modernization
- Multi-tenant SaaS Architecture
- Inventory & Fulfillment Systems
- Security-Critical Systems

Technical Skills

Languages: Python, JavaScript, TypeScript, C/C++, SQL

Backend & APIs: Flask, FastAPI, Connexion, SQLAlchemy, Pydantic, Celery, Redis, REST APIs, OpenAPI/Swagger, JWT, OAuth2, API integrations, background job orchestration, asynchronous processing pipelines

Frontend & UI Modernization: Vue 3 (Composition API), Nuxt, TypeScript, Pinia, Vite, Naive UI, legacy frontend modernization

Databases & Storage: PostgreSQL, MySQL, MSSQL, Redis, SQLite, S3-compatible object storage

Testing & Quality: pytest, unittest, Playwright, Selenium, CI automation

DevOps & Infrastructure: Docker, Linux, Bash, GitHub Actions, TeamCity, CI/CD workflows, containerized deployments

Security & Systems: Cryptography (AES-NI, RSA, ECC), TrustZone (QSEE / T-Base), kernel-mode & user-mode development

Architecture & Patterns: Service-oriented architecture, modular monoliths, repository pattern, DTO-based APIs, async processing workflows

Tools: Git, JIRA, Redmine, Perforce

Professional Experience

Commerix (E-commerce Automation Platform) — Senior Backend Engineer

Remote (Ukraine/USA) | Jan 2020 – Present

Tech: Python, Flask, FastAPI, SQLAlchemy, Pydantic, Celery, Redis, MySQL, PostgreSQL, Connexion/OpenAPI, Vue 3, TypeScript, Docker, pytest, Playwright

- Architected and developed a scalable automation platform for multi-channel e-commerce operations, supporting product lifecycle management, order orchestration, inventory synchronization, fulfillment workflows, and marketplace integrations.
- Designed and evolved modular backend architecture with clear service boundaries, repository patterns, DTO-based APIs, and asynchronous processing pipelines.
- Designed and implemented resilient integrations with Etsy, eBay, Shopify, WooCommerce, Wish, Joom, and 15+ supplier systems via REST APIs, XML feeds, and custom ingestion pipelines.
- Reduced manual operational workload by approximately 60% through automation of listings management, supplier imports, inventory synchronization, and fulfillment workflows.
- Implemented background processing using Celery and Redis for synchronization pipelines, imports, exports, scheduled jobs, and operational automation workflows.
- Designed versioned REST APIs using OpenAPI/Connexion to provide stable integration contracts between services and third-party systems.
- Implemented Redis-based caching and optimization strategies to reduce external API load and improve synchronization performance.
- Designed and implemented RBAC and multi-user collaboration capabilities with granular permissions and operational controls.
- Designed and implemented multi-warehouse inventory management, shipment orchestration, and shipping label integration workflows.
- Led modernization of a large legacy frontend from jQuery to Vue 3 + TypeScript architecture.
- Built operational tooling and internal admin interfaces for monitoring integrations, synchronization health, and automation workflows.

- Integrated LLM-based tooling for automated product content generation and metadata enrichment workflows.
- Drove architectural evolution toward a modern multi-tenant SaaS platform with improved modularity, operational isolation, and extensibility.

AlliedBits / Callback Technologies — C++ Developer

Remote | Ukraine | Dec 2017 – Jan 2020

Tech: C/C++, Windows SDK/WDK, Assembler, OpenSSL

- Developed and optimized a cross-platform cryptographic library with hardware-accelerated AES, SHA-2, and CRC32.
- Achieved up to 10× performance improvements using assembler optimizations and CPU intrinsics (AES-NI, SSE4.2, AVX2).
- Implemented performance-critical kernel-mode lookup and caching mechanisms, reducing rules-matching latency.
- Led development of a major API overhaul with multi-language bindings and improved build tooling..

Samsung R&D Institute Ukraine — Lead Engineer

Kyiv, Ukraine | Jun 2011 – Dec 2017

Tech: C/C++, Python, Boost, OpenSSL, XSLT, Tizen, TrustZone, Android NDK

- Developed security-critical TrustZone-based Android KeyMaster / KeyStore components for Samsung mobile devices, supporting secure cryptographic operations in trusted execution environments.
- Implemented ECC and variable-length RSA support, improving cryptographic flexibility and compatibility across multiple device generations.
- Designed modular cryptographic adapter layers supporting multiple crypto backends (OpenSSL, SCrypto, proprietary engines).
- Integrated and validated TrustZone support for new Exynos-based devices, improving hardware compatibility and release stability.
- Developed internal tooling and automation utilities for security validation, permission analysis, and report aggregation.
- Acted as Project Lead for Smart TV browser security add-on initiatives, coordinating a team of 4 engineers.
- Contributed to build system improvements and cross-team integration workflows for security-related components.

eCall LLC — C++ Developer

Kyiv/Chernihiv, Ukraine | Mar 2008 – Jun 2011

Tech: C++, wxWidgets, XML/XSLT, Lua, Cisco IPCC, MS SQL Server

- Developed configurable CTI and call-center productivity platforms with XML-driven workflow architecture.
- Designed reporting and analytics modules integrated with MS SQL Server and enterprise telephony systems.
- Contributed to migration from WTL to wxWidgets, improving maintainability and UI responsiveness.

AtomPark Software JSC — C++ Developer

Chernihiv, Ukraine | Dec 2006 – Mar 2008

Tech: C++, MFC, Windows DDK, WinSock, Wix Toolset, XSLT

- Developed kernel-mode network traffic monitoring components for HTTP, FTP, and messaging protocols.
- Built reporting infrastructure aggregating and transforming collected activity data into multiple export formats.

Education

M.S. in Software Engineering, Chernihiv National Technological University, Ukraine

B.S. in Computer Engineering, Chernihiv National Technological University, Ukraine